Mahi Singh

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Education

Vellore Institute of Technology — **Bhopal, India**
*Bachelor of Technology in Computer Science and Engineering — CGPA: **9.21/10.00*** — *Aug 2023 – May 2027*

Mercy Memorial School — **Uttar Pradesh, India**
*ISC (Class XII): **89.75%*** | *ICSE (Class X): **95.40%*** — *2023 & 2021*

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, HTML/CSS
AI & Machine Learning: TensorFlow, PyTorch, YOLOv11, Faster R-CNN, Computer Vision, Deep Learning, Google Gemini, Prompt Engineering, Pandas, NumPy
Web: Next.js, React, FastAPI, Node.js, Express.js, REST APIs, Tailwind CSS
Databases & Cloud: PostgreSQL, Supabase, MongoDB, MySQL, Vercel, Render, Railway
Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, CI/CD, Agile, SDLC
Spoken: English, Hindi

Experience

AI Engineer Intern — **Mar 2025 – Sep 2025**
AmasQIS.ai — *Oman (Remote)*

- Engineered a real-time PPE compliance detection system using Python, TensorFlow, and Faster R-CNN to process live multi-camera factory feeds, achieving **90+ precision**% on the validation set.
- Developed an end-to-end object detection pipeline to classify safety violations (helmets, vests, gloves) across concurrent camera streams for active factory floor monitoring.
- Optimized model inference for low-latency production deployment; collaborated with cross-functional engineering teams on iterative model improvements through structured optimization cycles.

Projects

JeevanTrack — AI Health Timeline Platform | ***Next.js, FastAPI, Supabase, Gemini 2.5, TypeScript*** Live App | GitHub

- Built an AI-powered health record platform that digitizes, summarizes, and visualizes patient medical history into a unified timeline using Gemini Vision, Next.js, FastAPI, and Supabase.
- Architected an offline-first data layer using localStorage-first caching and cloud synchronization, eliminating data fragmentation across **5+ core platform** features.
- Integrated Google Gemini 2.5 Flash via Vision API to automate medical report text extraction, reducing manual data entry by **80**% and rendering concise 1-page clinical summaries in under **30 seconds**.
- Designed a resilient multi-tier fallback system (local storage → rules-based matching → manual flow) ensuring zero user-facing failures and **100% reliability** during upstream AI disruptions.

knowYourFood — GenAI Ingredient Analyzer | ***React, FastAPI, Python, Gemini 2.0, PostgreSQL*** Live App | GitHub

- Developed a GenAI-powered food label analyzer that uses Google Gemini to explain ingredient safety, identify potentially harmful additives, and generate consumer-friendly health insights in real time.
- Architected the platform with a React frontend and asynchronous FastAPI backend, achieving a **13 second** average response time across **60+ product analyses**.
- Designed a secure REST API with structured prompt engineering, PostgreSQL authentication, and strict CORS security policies; implemented responsible AI output validation for safe, accurate responses.

KharpatNaashak — AI Weed Detection System | ***YOLOv11, PyTorch, OpenCV, Gradio, Python*** Model Space | GitHub

- Developed a computer vision system for automated weed detection in soybean fields using YOLOv11, enabling precise identification of crop weeds from field images.
- Trained a YOLOv11n model identifying 16 weed species, achieving **60.7% mAP@0.5**, **67.1% precision**, and 52.2% recall on the **MH-Weed16 dataset (6,656 images)**.
- Built a bilingual Gradio web interface with video upload, density heatmaps, and a herbicide cost-savings calculator demonstrating a **60–70%** chemical reduction through localized herbicide targeting.

Leadership & Volunteer

Technical & Community Volunteer — **Jul 2023 – May 2025**
eVidyaloka · Pahal Foundation · Pledge A Smile

- Taught mathematics for 2 years as a volunteer with eVidyaloka to 50+ students in rural Madhya Pradesh, improving assessment scores by 30%.
- Tutored construction workers' children of all ages through a volunteer teaching initiative (Pahal Foundation), helping some transition into formal schooling; also ran poster design and fundraising social campaigns for Pledge A Smile.